Artificial Intelligence and Machine Learning (CS52)

Term: Sep 2025 – Dec 2026
Unit 1 - Problems
Faculty Coordinator : Jamuna S Murthy

Write the PEAS description and task environment characteristics for:

1. Robot Vacuum Cleaner Operating in a Household Setting

- **Performance**: Clean floors effectively, avoid obstacles, conserve battery, cover maximum area.
- **Environment**: Household floors with various types (carpets, hardwood, etc.), obstacles (furniture, walls).
- **Actuators**: Wheels for movement, vacuum motor, rotating brushes.
- **Sensors**: Bump sensors, cliff sensors, dirt sensors, and infrared sensors for obstacle detection.

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2. Automated Medical Diagnostic System for Identifying Skin Cancer

- **Performance**: Accurate and early diagnosis, minimal false positives/negatives, user-friendly.
- **Environment**: Clinical or dermatological setting, patient's skin images.
- **Actuators**: Display screen for diagnosis output, possible printing/reporting system.
- **Sensors**: High-resolution camera for image capture, imaging sensors, skin imaging devices.

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3. Voice-Activated Personal Assistant (like Siri or Alexa) in a Smart Home

- **Performance**: Accurate responses to user commands, high speech recognition accuracy, response speed.
- **Environment**: Indoor settings, with potential background noise.
- **Actuators**: Speaker for responses, smart home device control interfaces.
- **Sensors**: Microphone array for voice detection, internet connectivity for information retrieval.

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4. Stock Trading Bot Designed to Maximize Returns on Stock Investments

- **Performance**: Maximize profits, minimize losses, respond to market trends.
- **Environment**: Online stock market data and trade platforms.
- **Actuators**: Trading platform interface for buying/selling stocks.
- **Sensors**: Financial data feeds, stock prices, economic indicators.

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5. Agricultural Robot Responsible for Monitoring and Optimizing Crop Growth in a Farm Field

- **Performance**: Monitor crop health, optimize watering and fertilization, detect pest infestations.
- **Environment**: Farm field with various crops and weather conditions.
- **Actuators**: Sprayers, wheels for mobility, arms for applying treatments.
- **Sensors**: Cameras, humidity sensors, soil moisture sensors, temperature sensors.

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6. Speech-Based Language Translation App for Real-Time Conversation Translation

- **Performance**: Accurate, fluent translation, minimal delay, high-quality audio output.
- **Environment**: Varied, with background noise, accents, and different languages.
- **Actuators**: Speaker or audio playback for translated speech.
- **Sensors**: Microphone for capturing spoken input, internet connectivity for database access.

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7. Search and Rescue Drone Designed for Locating People in a Disaster-Stricken Area

- **Performance**: Find people quickly, avoid hazards, navigate complex environments.
- **Environment**: Outdoor, potentially hazardous (rubble, debris, bad weather).
- **Actuators**: Propellers for movement, camera tilt/zoom.
- **Sensors**: Thermal imaging, GPS, high-resolution cameras, sonar for depth.

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8. Recommendation System that Suggests Movies or TV Shows to Users Based on Their Preferences

- **Performance**: Accurate and relevant recommendations, high user satisfaction.
- **Environment**: Online platform with user profiles and preferences.
- **Actuators**: Display on user's screen, notification systems.
- **Sensors**: User behavior tracking, rating history, browsing history.

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9. AI-Driven Chatbot for Customer Service That Assists Users with Common Questions and Issues

- **Performance**: Provide accurate answers, improve response time, handle multiple users.
- **Environment**: Online chat interface with diverse user queries.
- **Actuators**: Text output on chat interface.
- **Sensors**: Text input from users, database of FAQs, user information.

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10. Intelligent Traffic Management System That Controls Traffic Lights to Reduce Congestion in a City

- **Performance**: Reduce traffic congestion, prioritize emergency vehicles, minimize wait times.
- **Environment**: Urban traffic intersections, with varying vehicle and pedestrian flow.
- **Actuators**: Traffic lights, pedestrian signals.
- **Sensors**: Cameras, traffic flow sensors, GPS data from vehicles, traffic light status.

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11. Facial Recognition System Used for Security Purposes in Airports

- **Performance**: Accurately identify individuals, minimize false positives, comply with privacy laws.
- **Environment**: Airport terminals with varying lighting and large crowds.
- **Actuators**: Display or alert systems, access control gates.
- **Sensors**: High-resolution cameras, infrared for low-light detection.

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12. AI-Powered News Aggregator That Curates Personalized News Feeds for Users

- **Performance**: Provide relevant and timely news, high personalization.
- **Environment**: Online news sources and user profile database.
- **Actuators**: Display feed on user's device.
- **Sensors**: User interaction data, browsing history, news site APIs.

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13. Automated Plant-Watering System for Indoor Plants in a Greenhouse

- **Performance**: Maintain optimal soil moisture, reduce water wastage.
- **Environment**: Greenhouse with various types of plants and humidity levels.
- **Actuators**: Water pumps, irrigation system.
- **Sensors**: Soil moisture sensors, humidity sensors, temperature sensors.

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14. Real-Time Game-Playing AI Agent Designed to Play Chess Against Human Players

- **Performance**: High win rate, logical and adaptive strategy.
- **Environment**: Virtual chessboard with player interactions.
- **Actuators**: Move pieces on virtual board, display game status.
- **Sensors**: Input of opponent moves, game state.

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15. Self-Driving Car Navigating Urban Traffic

1. **Performance**: Safe and efficient driving, obey traffic laws, avoid obstacles.
2. **Environment**: Urban roads with dynamic traffic, pedestrians, and road signs.
3. **Actuators**: Steering, braking, acceleration, turn signals.
4. **Sensors**: Cameras, radar, lidar, GPS, proximity sensors.

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5. Given an initial state of a 8-puzzle problem and final state to be reached-

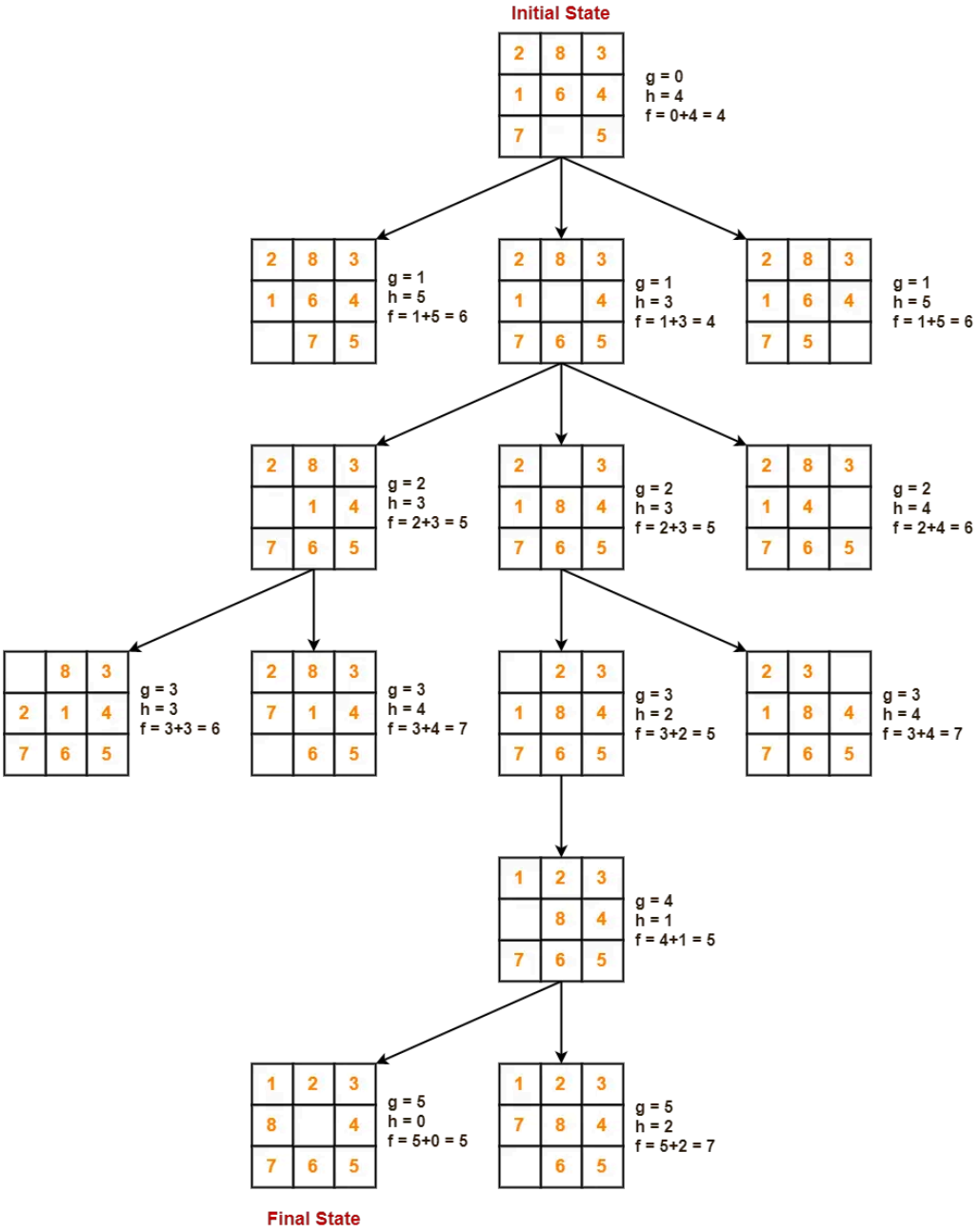
2	8	3
1	6	4
7		5

Initial State

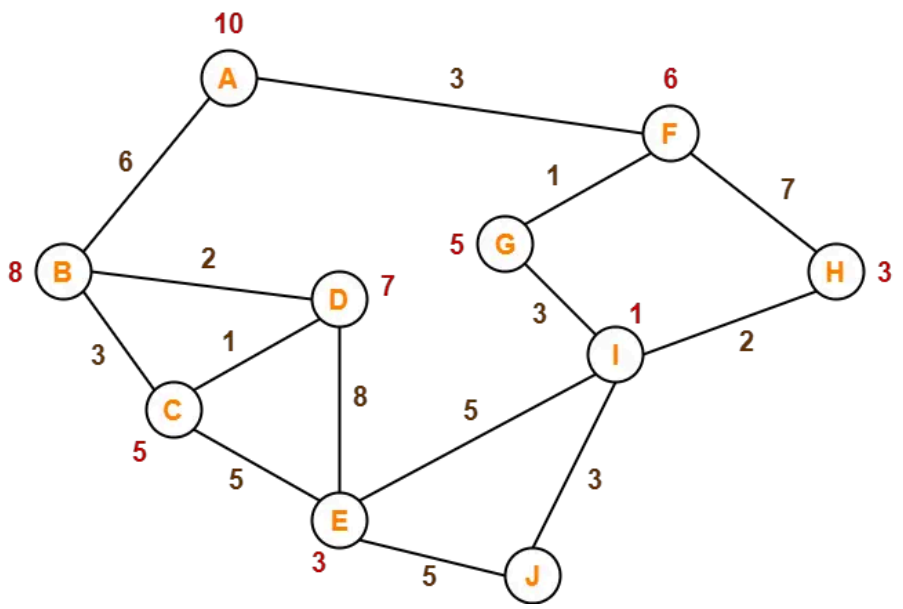
1	2	3
8		4
7	6	5

Final State

Find the most cost-effective path to reach the final state from initial state using A* Algorithm. Consider g(n) = Depth of node and h(n) = Number of misplaced tiles.



6. Consider the following graph. The numbers written on edges represent the distance between the nodes. The numbers written on nodes represent the heuristic value. Find the most cost-effective path to reach from start state A to final state J using A* Algorithm.



Solution

Step-01:

We start with node A.

- Node B and Node F can be reached from node A.

A* Algorithm calculates f(B) and f(F).

- f(B) = 6 + 8 = 14
- f(F) = 3 + 6 = 9

Since f(F) < f(B), so it decides to go to node F.

Path- A → F

Step-02:

Node G and Node H can be reached from node F.

A* Algorithm calculates f(G) and f(H).

6. f(G) = (3+1) + 5 = 9
7. f(H) = (3+7) + 3 = 13

Since f(G) < f(H), so it decides to go to node G.

Path- A → F → G

Step-03:

Node I can be reached from node G.

A* Algorithm calculates f(I).

f(I) = (3+1+3) + 1 = 8

It decides to go to node I.

Path- A → F → G → I

Step-04:

Node E, Node H and Node J can be reached from node I.

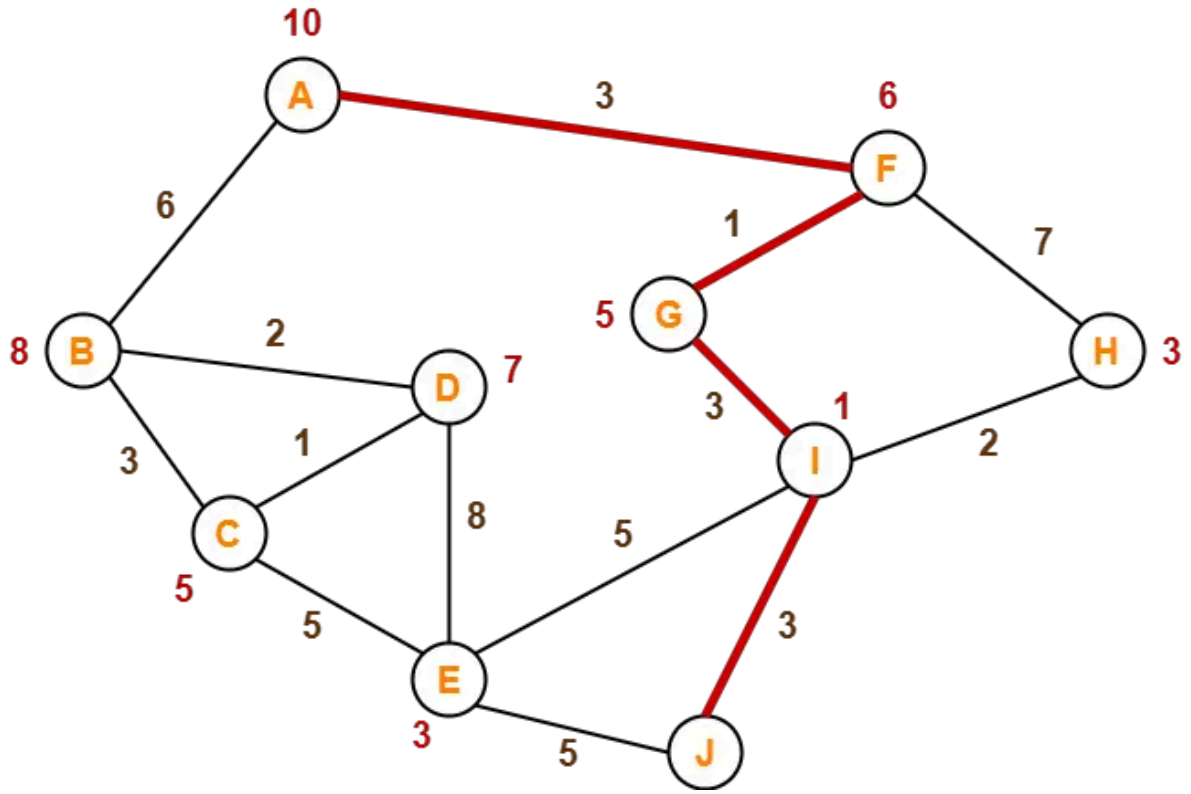
A* Algorithm calculates f(E), f(H) and f(J).

- f(E) = (3+1+3+5) + 3 = 15
- f(H) = (3+1+3+2) + 3 = 12
- f(J) = (3+1+3+3) + 0 = 10

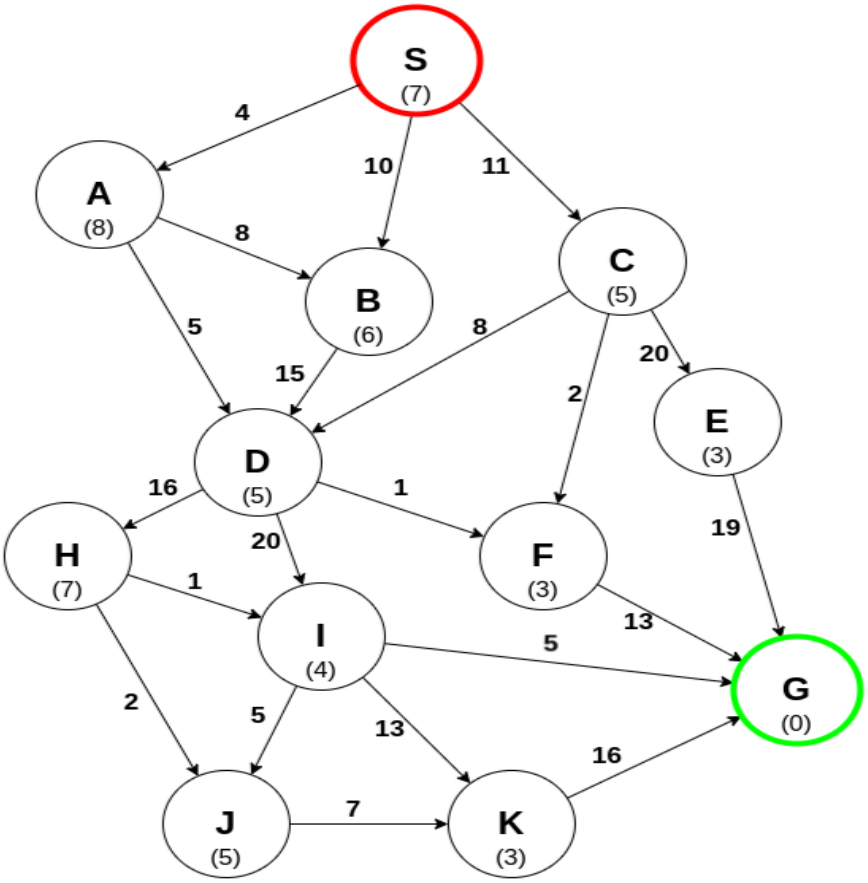
Since f(J) is least, so it decides to go to node J.

Path- A → F → G → I → J

This is the required shortest path from node A to node J.



7. Consider the following graph to find the shortest path from **S** to **G** using A* Algorithm:



Goal

Find the shortest path from **S** → **G** using the **A*** algorithm.

Heuristic (h) values

From the diagram (values inside parentheses):

Node	h(n)
S	7
A	8
B	6
C	5
D	5
E	3

F	3
G	0
H	7
I	4
J	5
K	3

📊 Edges (costs g)

From the diagram:

From	To	Cost
S	A	4
S	B	10
S	C	11
A	B	8
A	D	5
B	D	15
C	D	8
C	F	2
C	E	20
D	F	1
D	I	20
D	H	16
F	G	13
I	H	1
I	J	5
I	K	13
I	G	5
H	J	2

J	K	7
K	G	16
E	G	19

A* Algorithm Formula

For each node:

f(n)=g(n)+h(n)

where

- **`g(n)` = cost from start to node**
- **`h(n)` = heuristic (estimated cost to goal)**
- **`f(n)` = estimated total cost through node**

Step-by-Step Execution

Step 1: Start at S

```
Open list: [S]

g(S) = 0

f(S) = g + h = 0 + 7 = 7
```

Step 2: Expand S

Neighbors:

- A: g = 4 → f = 4 + 8 = 12
- B: g = 10 → f = 10 + 6 = 16
- C: g = 11 → f = 11 + 5 = 16

```
Open list: [A(12), B(16), C(16)]

Closed list: [S]
```

Choose **A** (lowest f).

Step 3: Expand A

Neighbors:

- B: g = 4 + 8 = 12 → f = 12 + 6 = 18 (worse than existing 16, ignore)
- D: g = 4 + 5 = 9 → f = 9 + 5 = 14

`Open: [D(14), B(16), C(16)]`

`Closed: [S, A]`

Choose **D (14)**.

Step 4: Expand D

Neighbors:

- F: g = 9 + 1 = 10 → f = 10 + 3 = 13
- I: g = 9 + 20 = 29 → f = 29 + 4 = 33
- H: g = 9 + 16 = 25 → f = 25 + 7 = 32

`Open: [F(13), B(16), C(16), I(33), H(32)]`

`Closed: [S, A, D]`

Choose **F (13)**.

Step 5: Expand F

Neighbors:

- G: g = 10 + 13 = 23 → f = 23 + 0 = 23

`Open: [G(23), B(16), C(16), H(32), I(33)]`

`Closed: [S, A, D, F]`

Choose **B (16)** (lowest f).

Step 6: Expand B

Neighbor:

- D: g = 10 + 15 = 25 → worse (ignore)

No better path → move on.

`Open: [C(16), G(23), H(32), I(33)]`

`Closed: [S, A, D, F, B]`

Choose **C (16)**.

Step 7: Expand C

Neighbors:

- D (already 9 < 19) → ignore
- F (already 10 < 13) → ignore
- E: g = 11 + 20 = 31 → f = 31 + 3 = 34

```
Open: [G(23), H(32), I(33), E(34)]
```

```
Closed: [S, A, D, F, B, C]
```

Choose **G (23) — Goal reached**

Final Path Reconstruction

Follow parent links back:

```
S → A → D → F → G
```

Step-by-Step Cost Trace

Edge	Cost
S → A	4
A → D	5
D → F	1
F → G	13
Total g(G)	**23**

Final Answer:

Metric	Value
Shortest Path	**S** → **A** → **D** → **F** → **G**
Total Cost	**23**
Sequence of expansions	S → A → D → F → B → C → G